

More realistic expressivity bounds for low-rank adaptation

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Introduction

LoRA – most popular yet understudied method of parameter-efficient fine-tuning

Problem

Existing theoretical estimates of LoRA expressive power is overly optimistic

Observation

Empirical results show that it is not the rank that limits the performance of the adapters

Proposal

Derive the more realistic expressive power bounds based on stable rank

Preliminary theory

Stable rank

For matrix A stable rank is $sr(A) = \frac{\|A\|_F^2}{\|A\|_{op}^2} = \frac{\sum_i \sigma_i^2(A)}{\sigma_1^2(A)}$

Lemma (single matrix approximation lower bound)

A – target matrix, $\hat{A}$ – an estimate. $S_A = sr(A)$, $S_{\hat{A}} = sr(\hat{A})$, $S_{\hat{A}} \leq S_A$ and $\sigma_1(\hat{A}) = \alpha \sigma_1(A)$, $\alpha \geq 0$ then:

$$\|A - \hat{A}\|_F \geq \sigma_1(A) \left(\frac{\sqrt{S_A} - \sqrt{S_{\hat{A}}}}{1 + \sqrt{S_{\hat{A}}}} \right)$$

Lemma (constructive lower bound)

Let A_R be the optimal unconstrained rank- R approximation of A with singular values $\sigma_1, \dots, \sigma_R$. $\overline{S_A} = sr(A_R)$. Then there exists $\hat{A}$ such that $\text{rank}(\hat{A}) \leq R$, $sr(\hat{A}) = S_{\hat{A}} \leq S \leq S_A$ and

$$\|A - \hat{A}\|_F^2 = \sum_{i=R+1}^D \sigma_i^2 + (1 - \gamma)^2 \sum_{i=2}^R \sigma_i^2, \quad \text{where } \gamma = \begin{cases} 1 & \text{if } \overline{S_A} \leq S \\ \sqrt{\frac{S-1}{\overline{S_A}-1}} & \text{if } \overline{S_A} > S \end{cases}$$

To get $\hat{A}$: SVD of $A_R = \sum_{i=1}^R \sigma_i u_i v_i^T$, $\sigma_2, \dots, \sigma_R \rightarrow \gamma \cdot \sigma_2, \dots, \gamma \cdot \sigma_R$

Fully connected network approximation

Notation

target FNN $\bar{f} := FNN_{\bar{L},D}(\cdot; (\bar{W}_l)_{l=1}^{\bar{L}}, (\bar{b}_l)_{l=1}^{\bar{L}})$,

frozen FNN $f_0 := FNN_{L,D}(\cdot; (W_l)_{l=1}^L, (b_l)_{l=1}^L)$,

adapted FNN $f := FNN_{L,D}(\cdot; (W_l + \Delta W_l)_{l=1}^L, (\hat{b}_l)_{l=1}^L)$

$b_l, \bar{b}_l, \hat{b}_l \in \mathbb{R}^D$, $W_l, \bar{W}_l, \Delta W_l \in \mathbb{R}^{D \times D}$, but $\Delta W_l = A_l B_l$, where $A_l \in \mathbb{R}^{D \times R}$ and $B_l \in \mathbb{R}^{R \times D}$ and $R \leq D$ – the LoRA rank. Additionally, we assume and $\bar{L} = L$.

Theorem (FNN error upper bound)

Let $x \in \mathbb{R}^d$ be the random input with covariance matrix $\Sigma = \mathbb{E}[xx^T]$ Let $E_l = \bar{W}_l - W_l$ be the weight discrepancy at layer l . Let $\sigma_{l,i}$ be the singular values of E_l . Assume $\hat{b}_l = \bar{b}_l$. Then there exist adapters ΔW_l with $\text{sr}(\Delta W_l) \leq S \ \forall l \in \overline{1, L}$ such that the expected error is bounded by:

$$\mathbb{E} \|f(x) - \bar{f}(x)\|_2 \leq \beta \sum_{l=1}^L \left(\prod_{k=l+1}^L \|\bar{W}_k\|_2 \right) \mathcal{T}_l, \text{ where}$$

$$\beta = \max_{i \in [L]} \left(\left(\prod_{j=1}^i \|\bar{W}_j\| \right) \sqrt{\text{tr}(\Sigma)} + \sum_{j=1}^i \left(\prod_{k=j+1}^i \|\bar{W}_k\| \right) \|\bar{b}_j\| \right)$$

$$\text{and } \mathcal{T}_l = \sqrt{\sum_{j=R+1}^D \sigma_{l,j}^2 + (1 - \gamma_l)^2 \sum_{j=2}^R \sigma_{l,j}^2}, \text{ with } \gamma_l = \min \left(1, \sqrt{\frac{S-1}{\text{sr}(E_l)-1}} \right)$$

Preliminary experiments

Controlled small scale setting

Randomly initialize 3-layer MLP with ReLU $\bar{f}_0$

Generate data $x \sim \mathcal{N}(0, I) \rightarrow \bar{f}_0 \rightarrow y$

Pretrain 3-layer MLP $\bar{f}_0$ to map x to y

Modify $\bar{f}_0 \rightarrow \bar{f}$, add $\Delta \bar{W}_2 \in \mathbb{R}^{d \times d}$ to second layer, with $\text{sr}(\Delta \bar{W}_2) \approx d = 256$

Train $\bar{f}_0 \rightarrow f$ with R -rank LoRA and measure $\text{sr}(\Delta W_2)$ and loss

R	4	16	64	256
SR	1.65	3.05	4.89	6.07
Loss	0.050	0.031	0.017	0.014

LLM and real downstream task

Pretrained Qwen2.5-0.5B model

Text to SQL dataset

Apply LoRA to each FFN block

Measure resulting loss and stable rank

R	4	16	64	256
SR	2.7	3.9	3.6	2.8
Loss	1.01	0.99	0.99	0.98

Moreover, full FT gives loss $\sim 0.7 \implies$ low SR is an actual problem

Conclusion

Current results

1. Fundamental limit of low stable rank approximation is shown
2. FNN error upper bound based on stable rank is proved
3. Optimization procedure low stable rank limitation is empirically demonstrated